

# Ben Irish

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## EDUCATION

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### University of California, Los Angeles

Los Angeles, CA

*Bachelor of Science in Statistics and Data Science, Minor in Systems Biology*

Expected Jun 2029

- Current GPA: 3.90/4.0
- Relevant coursework: Statistics 13 and 20, Math 33A (in progress), Life Science 7A, Environment 10, EPS SCI 15
- Transfer coursework (UMass Global): Calculus II, Calculus III (in progress)

## PROJECTS

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### Lyon and McAleer Creek Water Quality (*R*, Quarto)

Jul 2026 – Sep 2026

*For the Lake Forest Park Stewardship Foundation*

- Analyzed 30+ years of King County monitoring data for two urban creeks, identifying a ~1.4 °C step increase in summer temperature after 2013 and a separate, decades-long decline in dissolved oxygen
- Joined storm copper samples to NOAA precipitation records and tested whether the rainfall correlation held up when below-reporting-limit values were treated differently
- Delivered figures and a public report to the nonprofit, with a documented log of analysis decisions

### Breast Cancer Subtyping from Gene Expression (*R*, *limma*-*voom*)

Apr 2026 – Present

- Analyzed ~1,200 TCGA RNA-seq samples in R, reproducing the established four-subtype molecular classification via differential expression and unsupervised clustering
- Extended the analysis to test whether long non-coding RNAs recover the same subtype structure as protein-coding genes, quantified with adjusted Rand index
- Found lncRNAs recovered subtype structure as well as protein-coding genes

### Urban Coyotes in California (*R*, *sf*, Quarto)

May 2026 – Jun 2026

- Analyzed ~24,000 GBIF wildlife observation records in R, examining urban vs. non-urban patterns across geography, time, and season
- Found the urban share of observations roughly doubled from ~20% (2010) to ~50% (early 2020s), and separated real urban colonization from the growth of citizen-science apps as competing explanations
- Authored a reproducible report in Quarto, addressing the limits of citizen-science data

## EXPERIENCE

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### The Navigators – Doulos Leadership Program

Los Angeles, CA

*Program Member*

Sep 2026 – Present

- Selected for a year-long leadership development program with The Navigators
- Meet one-on-one with members to foster community; lead Bible study and worship

### Lakeside Bible Camp – Training in Christian Living (TCL)

Clinton, WA

*Cabin Leader*

Aug 2026

- Returned to lead a cabin of campers for the final week of the summer season

*Service Crew Member*

Jul 2025 – Aug 2025

- Served on a 20-person volunteer crew at a residential summer camp, supporting programming, food service, and facilities
- Built relationships with campers through service, encouragement, and intentional discipleship

### Wilderness Awareness School

Kenmore, WA

*Volunteer Camp Counselor*

Jun 2021 – Aug 2022

- Mentored and supervised campers ages 5–12, ensuring their safety, engagement, and wellbeing
- Instructed children in plant recognition, wilderness survival, and sustainable camping practices

## ADDITIONAL INFORMATION

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- Tools: R (tidyverse, sf, limma), Quarto, Git/GitHub
- Interests: Football (Seahawks fan; commissioner, 4 fantasy leagues), Guitar/Bass, Reading